# ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Minerals and Energy Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** Both A and R are true and R is the correct explanation of A. | Both true; R explains A |
| 2 | **(a)** Both A and R are true and R is the correct explanation of A. | Both true; R explains A |
| 3 | **(a)** aluminium | Light metal |
| 4 | **(a)** Both A and R are true and R is the correct explanation of A. | Both true; R explains A |
| 5 | **(a)** Digboi - oil | Oldest oil field |
| 6 | **(d)** using public transport | Less fuel use |
| 7 | **(a)** Both A and R are true and R is the correct explanation of A. | Both true; R explains A |
| 8 | **(d)** nuclear power plants | Atomic power |
| 9 | **(a)** Both A and R are true and R is the correct explanation of A. | The very high iron content makes it the finest ore; R explains A. |
| 10 | **(b)** Both A and R are true but R is not the correct explanation of A. | Both true uses of manganese, but R is an additional use, not an explanation. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | It gives large energy but has accident and waste risks (1); Safety and cost are debated (1). | 1+1 |
| 12 | Mining displaces tribals and destroys forests (1); It pollutes water and damages livelihoods (1). | 1+1 |
| 13 | Some regions get industry while others lack resources (1); It causes regional inequality and trade dependence (1). | 1+1 |
| 14 | Fossil fuels are limited and polluting (1); Renewable sources are cleaner and technology costs are falling (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Villages need fuel for cooking and light (1); Biogas and solar meet needs (1); This reduces deforestation (1). | 1+1+1 |
| 16 | Coal runs most thermal plants (1); It is cheap and abundant (1); But it pollutes and releases carbon dioxide (1). | 1+1+1 |
| 17 | Solar is available everywhere in sunny regions (1); Wind needs strong steady winds (1); Solar suits scattered villages, wind suits windy coasts (1). | 1+1+1 |
| | **OR:** Government sets policies and prices (1); It supports renewables and imports (1); It promotes saving and efficiency (1). | 1+1+1 |
| 18 | Mining gives raw materials and jobs (1); It damages forests and land (1); Rules, restoration and recycling balance both (1). | 1+1+1 |
| | **OR:** Scrap metal reused cuts mining (1); It saves energy and costs (1); It reduces pollution (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) Veins (1). (ii) Coal (or some iron ore) (1). (iii) Alluvial deposits of minerals in sands of valley floors and hill bases (1). (iv) Mineral resources are finite and non-renewable; they take millions of years to form, so conservation is essential (2). | 1+1+1+2 |
| 20 | Thermal power is the largest source, using coal, petroleum and natural gas, and plants are located near coalfields (1); Hydroelectricity uses falling water, is renewable and clean, but needs dams and suitable relief, as at Bhakra-Nangal (1); Nuclear power uses uranium and thorium, and stations are at Tarapur, Kota and Kalpakkam (1); Thermal power causes pollution, hydel power displaces people, and nuclear power carries safety and waste risks (1); A balanced mix is necessary to meet rising demand (1). | 1+1+1+1+1 |

| | | |
|---|---|---|
| | **OR:** Coal is the most abundant fossil fuel in India and is the main source of thermal power, but it pollutes the air (1); Petroleum from Mumbai High, Gujarat and Assam and natural gas from the Krishna-Godavari basin supply fuel, CNG and raw material for industries, but India imports much crude oil (1); Hydroelectricity, nuclear power from uranium and thorium, and non-conventional sources like solar, wind, biogas, tidal and geothermal are being developed (1); India has abundant sunshine, and the largest cluster of wind farms stretches from Nagercoil to Madurai (1); A shift to renewable sources, along with energy conservation, is required for sustainable supply (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Writing that **hematite** has more iron than magnetite; **magnetite** is richer (up to 70 per cent), though hematite is used more in quantity.
- Mixing the order of coal: peat, lignite, bituminous, **anthracite** (highest carbon, best quality).
- Classifying **hydel power** as non-conventional; hydel power, coal, petroleum, firewood and natural gas are conventional.
- Calling minerals renewable; they are exhaustible and take millions of years to form.
- Confusing **veins** and **lodes** (igneous and metamorphic rocks) with **placer deposits** (alluvial sands) and sedimentary beds.
- Forgetting the sources of nuclear minerals: **uranium and thorium** occur in Jharkhand and the Aravalli ranges of Rajasthan, and thorium is also found in the **monazite sands of Kerala**.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Minerals and Energy Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** biogas | Organic waste is used |
| 2 | **(c)** Mumbai High - coal | Mumbai High is an oilfield |
| 3 | **(a)** non-conventional energy sources are eco-friendly | Solar and wind are clean |
| 4 | **(c)** it is dangerous | Narrow tunnels |
| 5 | **(d)** making steel | Mica is an insulator |
| 6 | **(a)** Both A and R are true and R is the correct explanation of A. | Both true; R explains A |
| 7 | **(b)** India's import bill | India imports crude oil |
| 8 | **(a)** Both A and R are true and R is the correct explanation of A. | Both true; R explains A |
| 9 | **(c)** A is true but R is false. | A is true. Anthracite has the highest carbon content, so R is false. |
| 10 | **(d)** A is false but R is true. | Flowing water is replenished by the hydrological cycle, so hydel power is renewable; A is false, R is true. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | CNG burns cleaner than petrol or diesel (1); It reduces air pollution in cities (1). | 1+1 |
| 12 | Collecting and storing solar energy is costly (1); Dependence on weather and land needs limit scale (1). | 1+1 |
| 13 | Mining scars the land (1); Restoring land protects environment and livelihoods (1). | 1+1 |
| 14 | Exporting raw minerals depletes limited stocks (1); Value is gained by others if not processed at home (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Domestic crude oil is far less than demand (1); Transport and industry demand has grown (1); So India imports much oil (1). | 1+1+1 |
| 16 | Technology is costly and new (1); Research makes it efficient (1); Investment expands use (1). | 1+1+1 |
| 17 | Imported crude oil arrives by sea (1); Ports reduce transport costs (1); Refined products can be shipped out (1). | 1+1+1 |
| | **OR:** Better methods extract more from ore (1); Waste can be recycled (1); Low-grade ores become usable (1). | 1+1+1 |
| 18 | Coal burning releases carbon dioxide and ash (1); It causes air pollution and global warming (1); Mining also damages land (1). | 1+1+1 |
| | **OR:** Develop solar, wind and biogas (1); Improve efficiency (1); Use public transport (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) Biogas (1). (ii) Non-conventional (1). (iii) Energy for cooking or improved manure (1). (iv) Burning dung cakes and firewood wastes manure and reduces forests, whereas biogas is clean, efficient and still leaves manure for the fields (2). | 1+1+1+2 |
| 20 | Conventional sources such as coal, petroleum, natural gas and hydel power are widely used, but fossil fuels are exhaustible and pollute (1); Non-conventional sources such as solar, wind, tidal, geothermal and biogas are renewable and eco-friendly (1); Conventional sources have established technology and large capacity, while non-conventional ones are costlier to set up and site-specific (1); Examples: thermal plants near coalfields versus wind farms in Tamil Nadu and a solar plant at Bhuj (1); India needs both in the present but must increase non-conventional sources in the long run (1). | 1+1+1+1+1 |

| | | |
|---|---|---|
| | **OR:** Citizens can save electricity by switching off unused lights and using efficient appliances (1); They can use public transport and avoid wasteful use of fuel (1); They can use non-conventional sources such as solar heaters and biogas where possible (1); They can avoid wasting metal products and reuse or recycle scrap (1); Everyday small steps by many people reduce pressure on minerals and energy resources (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Writing that **hematite** has more iron than magnetite; **magnetite** is richer (up to 70 per cent), though hematite is used more in quantity.
- Mixing the order of coal: peat, lignite, bituminous, **anthracite** (highest carbon, best quality).
- Classifying **hydel power** as non-conventional; hydel power, coal, petroleum, firewood and natural gas are conventional.
- Calling minerals renewable; they are exhaustible and take millions of years to form.
- Confusing **veins** and **lodes** (igneous and metamorphic rocks) with **placer deposits** (alluvial sands) and sedimentary beds.
- Forgetting the sources of nuclear minerals: **uranium and thorium** occur in Jharkhand and the Aravalli ranges of Rajasthan, and thorium is also found in the **monazite sands of Kerala**.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Minerals and Energy Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** the Gulf of Khambhat and Gulf of Kuchchh | Coastal inlets |
| 2 | **(b)** Japan and South Korea | Through ports |
| 3 | **(b)** thermal power | Coal-based power |
| 4 | **(b)** switching off unused lights | Reduces demand |
| 5 | **(a)** it needs strong and steady winds | Site specific |
| 6 | **(a)** coal | A fossil fuel |
| 7 | **(a)** Both A and R are true and R is the correct explanation of A. | Both true; R explains A |
| 8 | **(a)** thermal power generation | Brown coal |
| 9 | **(a)** Both A and R are true and R is the correct explanation of A. | These minerals resist corrosion and so accumulate in sand deposits; R explains A. |
| 10 | **(b)** Both A and R are true but R is not the correct explanation of A. | Both true statements about energy resources; they are unrelated. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Dams displace people and submerge forests (1); They change river flow and cause conflicts (1). | 1+1 |
| 12 | Factories stop or slow production (1); Losses and job cuts follow (1). | 1+1 |
| 13 | Saving energy reduces demand and imports (1); It conserves resources and the environment (1). | 1+1 |
| 14 | They are inexhaustible and cause less pollution (1); They reduce dependence on fossil fuel imports (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Tidal energy needs special coastal sites (1); Geothermal needs hot rock zones (1); Both are costly and limited in India (1). | 1+1+1 |
| 16 | Mumbai High is offshore and the largest producer (1); Digboi in Assam is the oldest onshore field (1); Mumbai High needs costly rigs (1). | 1+1+1 |
| 17 | Mineral-rich areas get industries and jobs (1); But some, like Jharkhand, remain poor (1); Local people need fair benefits (1). | 1+1+1 |
| | **OR:** Nuclear plants have strict safety systems (1); But accidents and waste disposal are serious risks (1); Safety needs constant care (1). | 1+1+1 |
| 18 | Industries use iron, copper, aluminium, limestone and fuel (1); Without minerals, no steel, machinery or cement (1); So minerals support industrial growth (1). | 1+1+1 |
| | **OR:** More vehicles raise fuel demand (1); Road transport depends on oil (1); Public transport and CNG reduce use (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) Anthracite (1). (ii) Neyveli in Tamil Nadu (1). (iii) Bituminous coal (1). (iv) Lignite has high moisture and low carbon content, hence low heating capacity, while anthracite has high carbon content (2). | 1+1+1+2 |
| 20 | India depends heavily on imported crude oil, so oil price rises affect the economy (1); Demand for electricity and fuel is rising rapidly with industrialisation and transport (1); Coal is abundant but polluting, and hydel and nuclear power have limits (1); Renewable sources need investment and technology (1); Security requires diversification of sources, conservation and efficient use (1). | 1+1+1+1+1 |

| | | |
|---|---|---|
| | **OR:** Coal is concentrated in the Gondwana fields of eastern India, so thermal power and steel industries cluster there (1); Oil comes from Mumbai High, Assam and Gujarat, with refineries on the coast and in Assam (1); Large oil imports shape the economy and the balance of payments (1); Natural gas pipelines like the Hazira-Vijaipur-Jagdishpur line link fields to fertiliser and power plants (1); The geography therefore influences industrial location, trade and regional development (1). | 1+1+1+1+1 |

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- Classifying **hydel power** as non-conventional; hydel power, coal, petroleum, firewood and natural gas are conventional.
- Calling minerals renewable; they are exhaustible and take millions of years to form.
- Confusing **veins** and **lodes** (igneous and metamorphic rocks) with **placer deposits** (alluvial sands) and sedimentary beds.
- Forgetting the sources of nuclear minerals: **uranium and thorium** occur in Jharkhand and the Aravalli ranges of Rajasthan, and thorium is also found in the **monazite sands of Kerala**.

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